

Bhupesh Bandi

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Professional Summary

- Experienced **Generative AI Engineer** with 5+ years of experience designing and deploying **LLM-based and AI systems**, leveraging **LangChain, LangGraph, and AutoGen** for multi-agent orchestration and autonomous task execution.
- Proficient in developing **RAG architectures** using **GPT-4, T5, FAISS, and Pinecone**, enabling context-grounded reasoning, tool use, and multi-step decision-making.
- Skilled in building **low-latency AI microservices** using **FastAPI** and **asynchronous pipelines**, integrated with **Docker, Kubernetes, and Terraform** for scalable cloud deployment.
- Hands-on experience in **AI workflow automation** using **AWS Bedrock, Vertex AI, and Lambda**, streamlining prompt routing, model orchestration, and inference management.
- Deep understanding of **prompt engineering, few-shot learning, and instruction tuning** for improving model reasoning, factual accuracy, and coherence across generative tasks.
- Strong background in **deep learning and NLP** using **TensorFlow, PyTorch, and Keras**, with experience in implementing architectures like **CNNs, RNNs, GRUs, and LSTMs**.
- Expertise in **data engineering** — building robust **ETL pipelines** using **Pandas, NumPy, SQL, Apache Airflow, and Spark** to support scalable AI and ML workloads.
- Applied **MLOps principles** using **AWS SageMaker, Vertex AI, MLflow, and CI/CD** automation tools to streamline training, testing, versioning, and deployment of models.
- Proficient in **vector database management, semantic search, and embedding optimization**, enabling high-performance retrieval and context-driven AI applications.
- Experienced in **real-time data processing** with **Kafka and Spark Streaming**, building event-driven inference and continuous learning pipelines for adaptive AI systems.
- Adept at building **visualization and monitoring tools** using **Power BI, Grafana, and Prometheus** for tracking performance, model drift, and business insights.
- Strong focus on **secure, compliant, and scalable AI system design**, implementing **IAM-based access control, container security, and cloud governance** across hybrid environments.

Technical Skills:

Languages & Frameworks: Python, SQL, JavaScript, FastAPI, Django REST Framework, Flask, Shell Scripting.

AI / ML / GenAI: TensorFlow, PyTorch, Keras, Scikit-Learn, Hugging Face Transformers, BERT, RoBERTa, T5, GPT-4, LangChain, LangGraph, AutoGen, Vertex AI, AWS Bedrock, OpenAI APIs, RAG Pipelines, Prompt Engineering, Few-Shot Learning, Instruction Tuning, Transfer Learning, Solution Architecture Validation, AI System Design.

Data Engineering & Processing: Pandas, NumPy, Apache Spark, Apache Airflow, Kafka, MLflow, Feature Engineering, ETL Pipelines, Data Cleaning, Transformation, SQL-based Data Modeling, Embedding Optimization, Semantic Search, FAISS, Pinecone.

Cloud, MLOps & Infrastructure: AWS (SageMaker, Lambda, Glue, Bedrock, S3, Redshift, IAM), GCP (Vertex AI, BigQuery, Storage, Compute Engine), Azure ML, Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD, Prometheus, Grafana.

Databases & Visualization: PostgreSQL, MySQL, MongoDB, DynamoDB, Snowflake, Power BI, Tableau, Grafana, Matplotlib, Seaborn.

Version Control & Collaboration: Git, GitHub, Bitbucket, JIRA, Confluence.

Professional Experience

Broadcom

Jun 2024 – Current

Gen AI Developer

Kansas City, MO

- Designed and implemented **LLM-powered applications** using **LangChain, LangGraph, and AutoGen** frameworks to enable intelligent multi-agent collaboration, allowing models to interact dynamically and complete complex, multi-step tasks autonomously.
- Built advanced **Retrieval-Augmented Generation (RAG) pipelines** integrating **GPT-4** and **T5** models with **FAISS** and **Pinecone** vector databases, ensuring accurate retrieval of relevant context for enhanced generative reasoning and factual consistency.
- Designed and built **AI and LLM solutions** from **scratch**, owning the **full lifecycle** from ideation and POC through production deployment and ongoing support.
- Developed **low-latency AI microservices** using **FastAPI** and **asynchronous inference pipelines**, improving performance and scalability while deploying containerized applications via **Docker, Kubernetes, and Terraform** for infrastructure automation.
- Created end-to-end AI orchestration systems using **AWS Bedrock, Vertex AI, and Lambda** workflows to automate prompt routing, dynamic tool invocation, and multi-agent coordination, enabling seamless integration of AI-driven decision processes.
- Documented **solution architectures, design decisions, trade-offs**, and lessons learned to support knowledge sharing and future system scalability.

- Applied **prompt engineering**, **few-shot learning**, and **instruction-tuning** strategies to optimize generative AI models for context retention, precision, and coherence across various conversational and reasoning-based tasks.
- Engineered **data ingestion and preprocessing pipelines** in **Python** using **Pandas**, **NumPy**, and **Scikit-learn**, ensuring clean, structured, and balanced datasets for robust machine learning and natural language processing workflows.
- Monitored system and model performance using **Prometheus**, collecting detailed metrics to track drift and resource usage, and visualized the insights using **Grafana** dashboards to support proactive decision-making.
- Developed **deep learning and NLP models** with **TensorFlow**, **PyTorch**, and **Keras**, utilizing architectures like **CNNs**, **RNNs**, **GRUs**, and **LSTMs** for predictive modeling, language understanding, and sentiment analysis.
- Implemented **GenAIOps** and **MLOps** pipelines using **Vertex AI** and **SageMaker**, supporting continuous deployment, monitoring, and production support of LLM systems.
- Designed interactive **Power BI** dashboards and performance monitoring tools to visualize model behavior, performance metrics, and real-time insights, enabling data-driven decision-making and continuous improvement.
- Ensured **secure, scalable AI system deployment** across cloud and hybrid environments using **IAM-based access control** and **data governance best practices**, maintaining compliance, reliability, and long-term maintainability of AI infrastructure.

Saint Louis University AI/ML Engineer

Mar 2023 – May 2024
Saint Louis, MO

- Worked with **BERT**, **RoBERTa**, and **T5** to build intelligent AI features, using each model's strengths for tasks like summarization, semantic understanding, extraction, and conversational flows, ensuring output is meaningful and contextually aligned.
- Used **LangChain**, **LangGraph**, and **AutoGen** to create smart multi-agent workflows where different agents can reason, retrieve data, and call tools, allowing automation of complex multi-step tasks that previously required manual coordination.
- Led innovation-driven POCs to evaluate new GenAI capabilities, **agent architectures**, and **orchestration patterns**, accelerating experimentation into production-ready solutions.
- Designed efficient RAG pipelines using **FAISS** and **Pinecone**, managing vector stores and embeddings to enable highly accurate semantic search, contextual retrieval, and grounded LLM responses that prioritize relevance over guessing.
- Partnered with **product leaders** and **stakeholders** to explain **AI capabilities**, **limitations**, and trade-offs, supporting informed decision-making and roadmap planning.
- Practiced advanced prompt engineering by crafting **few-shot examples**, persona-driven prompts, and structured reasoning templates, improving clarity, stability, and accuracy of LLM outputs across a wide range of scenarios.
- Developed asynchronous backend services with **FastAPI** and **Python**, enabling real-time inference, low-latency streaming responses, and seamless integration with AI pipelines that process text, embeddings, and model outputs.
- Used **TensorFlow**, **PyTorch**, and **Scikit-Learn** for experimentation and model development, from implementing deep learning architectures to building predictive models tailored to various use cases and data characteristics.
- Performed robust data preprocessing using **Pandas**, **NumPy**, and **SQL**, handling cleaning, transformation, feature extraction, and dataset structuring to ensure consistent, high-quality inputs for AI and ML systems.
- Deployed AI microservices using **Docker** and **Kubernetes**, working with containerization and orchestration to ensure scalable, resilient, and maintainable deployment environments for production AI applications.
- Integrated workflows with **AWS**, **GCP**, and **Vertex AI**, using cloud-native tools to streamline model deployment, storage, pipeline automation, and scaling AI systems based on workload demands.
- Built evaluation and monitoring frameworks through automated scoring, output validation, and A/B testing, enabling continuous improvement of model accuracy, latency, and reliability across different versions of prompts and pipelines.
- Applied strong security and compliance principles with **IAM**, logging, and role-based controls, ensuring that AI services are secure, monitored, and safe to operate in production environments with sensitive data.

Tata Consultancy Services AI/Data Engineer

Dec 2021 - Jan 2023
Bangalore, India

- Worked on designing **AI and ML** components that fit cleanly into existing system architectures, ensuring smooth communication between data pipelines, application layers, and deployed models.
- Built text-processing workflows using early **LLM models**, improving their accuracy by combining them with retrieval and feature-based ML steps so outputs were grounded in relevant information rather than generic responses.
- Developed backend services in **Python**, using frameworks like **Django REST Framework** and **FastAPI** to expose ML and AI inference endpoints in a reliable and scalable manner.
- Created automated data pipelines with **AWS Glue**, relied on **Amazon S3** for storing curated datasets, and used **Redshift** to support analytical workloads and model-ready transformations.
- Integrated scalable data platforms such as **Snowflake** with cloud data layers, enabling secure and performant access for ML experimentation and iterative model development.
- Designed multi-step AI workflows by combining LLM reasoning with conventional ML techniques such as regression, clustering, and embedding-based retrieval, improving contextual understanding and the overall quality of outputs.
- Improved observability in AI/ML pipelines by implementing structured logging, evaluation checkpoints, and transparent tracking of model inputs and outputs, making debugging and model iteration more effective.

- Performed feature engineering using **Pandas**, **NumPy**, and **SQL**, preparing consistent datasets for ML models including classification, anomaly detection, and clustering tasks.
- Developed ML components using **TensorFlow**, **PyTorch**, and **Scikit-Learn**, applying them to build and evaluate models such as Random Forests, logistic regression, and basic neural network architectures depending on the use case.
- Contributed to CI/CD processes using **Jenkins**, helping automate testing, packaging, and deployment workflows so model updates and pipeline enhancements moved safely into production.
- Built monitoring dashboards using cloud-native streaming tools and visualization platforms to track model performance, data flow reliability, and system health in real time, supporting stable ongoing operations.

Cyient
Machine Learning Engineer

Aug 2020 – Nov 2021
Hyderabad, India

- Designed and deployed machine learning models using **Python**, working with frameworks like **TensorFlow** and **PyTorch** to build both classical ML models and deep learning architectures suited for various predictive tasks.
- Developed automated training pipelines using **Apache Airflow**, leveraging its workflow orchestration capabilities to manage data ingestion and feature engineering, while using **MLflow** to track experiments, metrics, and model versions.
- Processed large datasets using **SQL** for querying, **Pandas** for detailed data manipulation, and **Apache Spark** for distributed computation, ensuring efficient and scalable feature preparation.
- Built ETL workflows with **Apache Spark**, applying its distributed processing capabilities to cleanse, transform, and standardize incoming data before loading it into appropriate storage or analytics environments.
- Containerized ML applications using **Docker** and deployed them on **Kubernetes**, taking advantage of Kubernetes features such as autoscaling, rolling updates, and cluster-level monitoring to maintain stable production systems.
- Implemented real-time inference pipelines using **Kafka**, utilizing its event streaming strengths to capture continuous data flow, and enhanced these pipelines with **Spark Streaming** when additional computation was required.
- Developed deep learning solutions using architectures such as CNNs, RNNs, and Autoencoders, applying the capabilities of **TensorFlow**, **PyTorch**, and **OpenCV** to handle image-based, sequential, and unstructured data processing tasks.
- Applied classical and unsupervised machine learning techniques including Random Forests, Gradient Boosting, K-Means, and DBSCAN, selecting appropriate algorithms based on interpretability, scalability, and suitability for the dataset.
- Automated CI/CD workflows using **GitLab CI** and **Jenkins**, using these tools to enforce version control, automate testing, and streamline deployment steps, ensuring a smooth transition from development to production.

Educational Details:

Master of Science (M.S.) in Information Systems

- Saint Louis University, St. Louis MO, USA | May 2024

Bachelor of Technology (B.Tech.) in Mechanical Engineering

- Siddhartha Institute of Engineering & Technology, Hyderabad, India

Certifications:

- Microsoft Azure Fundamentals (AZ-900)
- Microsoft Power Platform Fundamentals (PL-900)
- Databricks Lakehouse Fundamentals
- Google IT Support